

thus the complete soln is

$$\phi(x^2 + y^2 + z^2, lx + my + nz) = 0 \text{ Ans.}$$

Solve

-mn

$$\textcircled{1} \quad 2p + 5q = 1 \quad \left(p = \frac{\partial z}{\partial x}, \quad q = \frac{\partial z}{\partial y} \right)$$

$$2) \quad y^2 p + x^2 q = x^2 y^2 z^2$$

$$3) \quad \frac{y^2 z}{x} p + xz q = y^2$$

$$4) \quad x(y^2 + z) \frac{\partial z}{\partial x} - y(x^2 + z) \frac{\partial z}{\partial y} = z(x^2 - y^2)$$

$$5) (y^2 + z^2) \frac{\partial z}{\partial x} - xy \frac{\partial z}{\partial y} + xz = 0$$

$$6) p \tan x + q \tan y = \tan z$$

$$7) x^2 \frac{\partial z}{\partial x} + y^2 \frac{\partial z}{\partial y} = z^2$$

$$8) x^2(y-z)p + y^2(z-x)q = z^2(x-y)$$

$$9) x(y^2 - z^2)p + y(z^2 - x^2)q = z(x^2 - y^2)$$

$$10) x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = z.$$

Charpit's Method :-

Charpit's method is a general method for finding the complete integral of a non-linear Partial diff eqn.

working rules for using charpit's method :-

Step (I) Shift all the terms of the given eqn to the left side and denote the left side by $f(x, y, z, p, q)$.

Step (II) :- Find $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$, $\frac{\partial f}{\partial z}$, $\frac{\partial f}{\partial p}$ and $\frac{\partial f}{\partial q}$.

Step (III) :- write the charpit's auxiliary eqn, & putting the values of Partial derivatives of f & simplify.

charpit's auxiliary eqn is

$$\Rightarrow \frac{dp}{\frac{\partial f}{\partial x} + p \frac{\partial f}{\partial z}} = \frac{dq}{\frac{\partial f}{\partial y} + q \frac{\partial f}{\partial z}} = \frac{dz}{-p \frac{\partial f}{\partial p} - q \frac{\partial f}{\partial q}}$$

$$= \frac{dx}{-\frac{\partial f}{\partial p}} = \frac{dy}{-\frac{\partial f}{\partial q}}$$

Step (IV):- Select any two fractions so that the resulting integral is the simplest ~~fraction~~ relation involving at least one of p & q . This relation and the given equation are solved to find the values of p & q .

Step (V):- Put the values of p & q in the eqn $dz = p dx + q dy$ and integrate. This gives the complete soln of the given eqn.

① Q Find the complete soln by Charpit's method

$$Z = px + qy + p^2 + q^2$$

Sol let $f(x, y, z, p, q) = Z - px - qy - p^2 - q^2$

$$\frac{\partial f}{\partial x} = -p, \quad \frac{\partial f}{\partial y} = -q, \quad \frac{\partial f}{\partial z} = 1,$$

$$\frac{\partial f}{\partial p} = -x - 2p, \quad \frac{\partial f}{\partial q} = -y - 2q.$$

Charpit's auxiliary eqn is

$$\begin{aligned} \frac{dp}{\frac{\partial f}{\partial x} + p \frac{\partial f}{\partial z}} &= \frac{dq}{\frac{\partial f}{\partial y} + q \frac{\partial f}{\partial z}} = \frac{dz}{-p \frac{\partial f}{\partial p} - q \frac{\partial f}{\partial q}} \\ &= \frac{dx}{-\frac{\partial f}{\partial p}} = \frac{dy}{-\frac{\partial f}{\partial q}} \end{aligned}$$

$$\begin{aligned} \Rightarrow \frac{dp}{-p + p \cdot 1} &= \frac{dq}{-q + q \cdot 1} = \frac{dz}{-p \cdot (-x - 2p) - q \cdot (-y - 2q)} \\ &= \frac{dx}{-(-x - 2p)} = \frac{dy}{-(-y - 2q)} \end{aligned}$$

$$\Rightarrow \frac{dp}{0} = \frac{dq}{0} = \frac{dz}{px + qy + 2p^2 + 2q^2} =$$

$$= \frac{dx}{-x-2p} = \frac{dy}{-y-2q}$$

$$\Rightarrow dp = 0 \Rightarrow \text{let } p = a$$

$$\text{Similarly } dq = 0 \Rightarrow q = b$$

Putting the value of p & q
in the eqⁿ

$$dz = p dx + q dy$$

$$\Rightarrow dz = a dx + b dy$$

$$\text{integrating } Z = ax + by + c \text{ Ans.}$$

2) Solve

$$2xz - bx^2 - 2qxy + bq = 0$$

Sol Let $f(x, y, z, b, q) =$

$$2xz - bx^2 - 2qxy + bq = 0$$

Charpit's auxiliary eqn is

$$\frac{dp}{\frac{\partial f}{\partial x} + p \frac{\partial f}{\partial z}} = \frac{dq}{\frac{\partial f}{\partial y} + q \frac{\partial f}{\partial z}} = \frac{dz}{-p \frac{\partial f}{\partial b} - q \frac{\partial f}{\partial q}}$$
$$= \frac{dx}{-\frac{\partial f}{\partial b}} = \frac{dy}{-\frac{\partial f}{\partial q}} \quad \text{--- (1)}$$

$$\Rightarrow \frac{\partial f}{\partial x} = \frac{2z - 2qx}{-2bx}, \quad \frac{\partial f}{\partial y} = -2qx$$

$$\frac{\partial f}{\partial z} = 2x, \quad \frac{\partial f}{\partial b} = -x^2 + q, \quad \frac{\partial f}{\partial q} = -2xy + b$$